Box Office Intelligence:

Exploratory Data Analysis for a New Movie Studio

Data Sources: bom.movie | tn.movie | rt.reviews | rt.movie

2010–2018 | 3,359 Films | 5 Datasets

Presented by Group 1 · Data Science Cohort · Moringa School

- Chelimo Ivy
- Shadrack Symekah
- Richard Oketch
- Orandi Felix

Business Understanding

Setting the stage — why movies, why now?

The global box office averaged ~$28B per year from 2010–2018, with over 400 films released annually.

The Opportunity

- Tech companies are investing billions in content
- Streaming has expanded global audience reach
- Data now enables smarter greenlight decisions

The Challenge

- Most films lose money — failure rate is high
- Hit-driven industry with uncertain returns
- Require strategic data analysis to compete

Our Goal

- Identify genres, budgets & quality thresholds
- Build a profitable, sustainable slate strategy
- Enter with evidence-backed investment decisions

Data Understanding

Where did the data come from, and what does it tell us?

BOM

bom.movie

Studio names, domestic gross, and release year

3,359 films

TN

tn.movie

Production budgets and worldwide gross figures

5,415 films

RT

rt.reviews

Genre tags and release metadata for each title

Genre data

RT

rt.movie

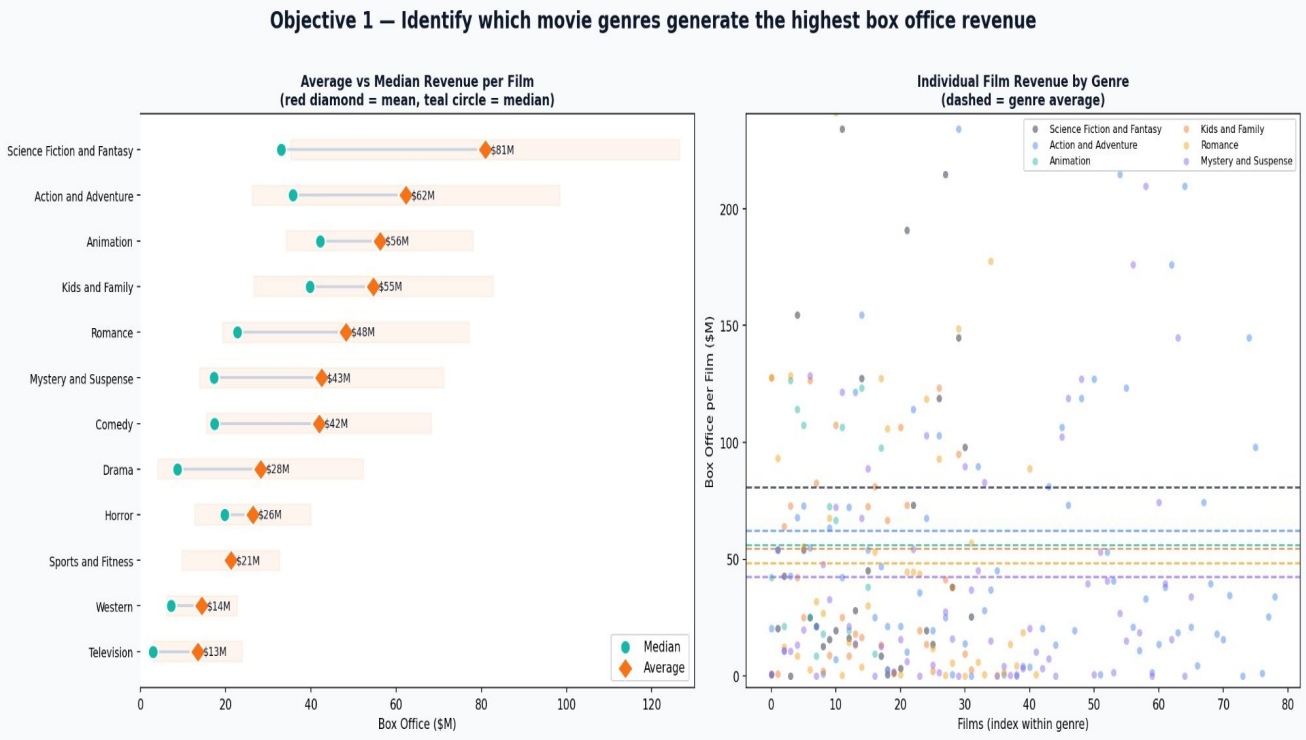
Critic scores (Fresh %) and total review counts

Scores + reviews

Data Preparation: All datasets were cleaned to remove missing values, normalize currency figures, and merged on shared film titles. Analysis covers theatrical films released 2010–2018.

Analysis 1 — Genre Revenue

Which genres consistently earn the most at the box office?



Sci-Fi & Fantasy

$81M avg

#1 earning genre

Action & Adventure

$62M avg

#2 earning genre

Animation

$56M avg

#3 earning genre

Finding 1 — Genre Revenue

What the data tells us about genre performance

- **Sci-Fi & Fantasy** leads at $80.9M avg — a few mega-hits pull the mean up, but the median is also strong.
- **Action & Adventure** offers the best volume + performance balance: 366 films at $62.3M avg.
- **Animation** has only 47 films but is the most consistent — almost no animated wide releases failed.
- **Comedy & Drama** dominate total volume but earn only $41.9M and $28.3M avg — inefficient for a new studio.
- **Horror** earns just $26.4M avg but has exceptional ROI potential at micro-budgets ($1–5M).

Key Insight — Genre Strategy

Strategic implication of the genre analysis

Genre choice significantly influences revenue potential, but the optimal strategy depends on balancing earning power, consistency, and scalability.

High earnings

Sci-Fi, Fantasy & Action deliver the highest avg revenue per film

Consistency

Animation has the lowest failure rate — nearly every wide release succeeds

Caution

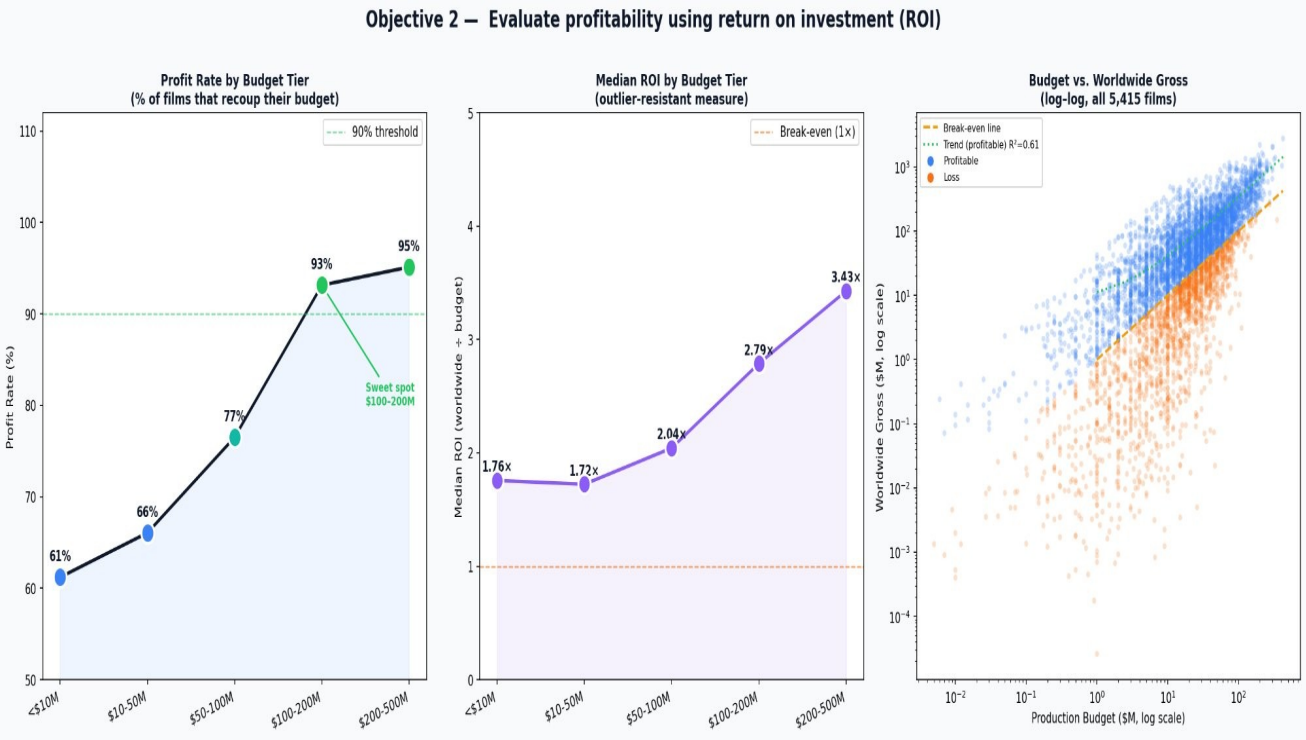
Comedy & Drama require high volume to generate impact — too risky to lead with

Niche play

Horror at micro-budgets ($1–5M) offers strong ROI for cash-flow diversification

Analysis 2 — Budget & ROI

How does production budget affect profitability and return on investment?



Sweet Spot Budget

$100–200M

93% profit rate

Median ROI at that tier

2.79×

Return on investment

Large budgets ($200M+)

3.43×

ROI — 95% profit rate

Finding 2 — Budget & ROI

How production budget size relates to profitability

- **$100M–$200M tier** achieves a 93.1% profit rate — 93 of every 100 films recoup their budget.
- **Median ROI of 2.79×** means a $150M film earns ~$418M worldwide at the median.
- **Micro & low tiers (<$50M)** have only 61–66% profit rates — cheap films carry more risk than assumed.
- **High tier ($200M+)** has the best stats (95.1%, 3.43× ROI) but requires established infrastructure.
- **Log-log scatter** shows a clear pattern: bigger budgets cluster near break-even; small budgets widely dispersed below.

Key Insight — Budget Strategy

Strategic implication of the budget & ROI analysis

The optimal strategy is not to minimize cost or maximize scale — there is a 'sweet spot' in the $100M–$200M range where profitability and risk are best balanced.

< $50M

Profit Rate

61–66%

Median ROI

1.72–1.76×

High variability
Avoid as lead tier

$100M–$200M

Profit Rate

93.1%

Median ROI

2.79×

Sweet spot
Start here

$200M+

Profit Rate

95.1%

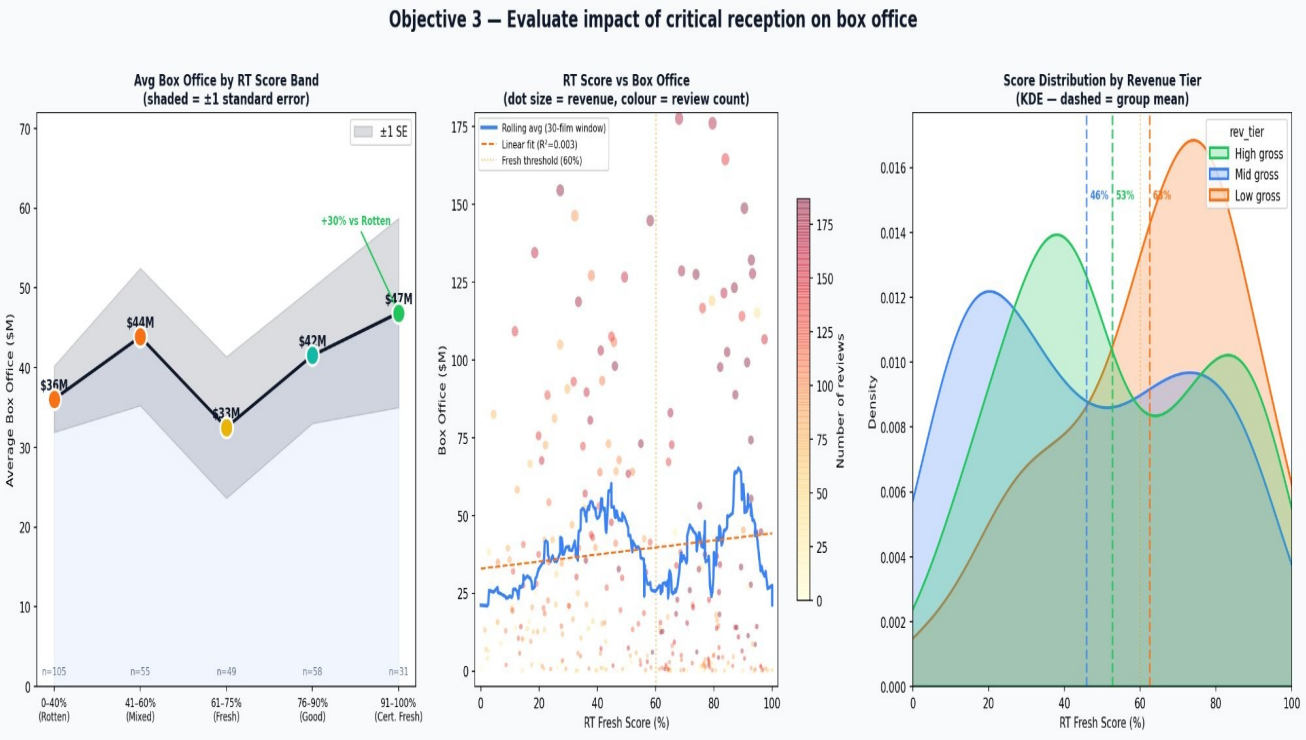
Median ROI

3.43×

Best returns
Scale into later

Analysis 3 — Critical Reception

Does a good Rotten Tomatoes score actually drive more box office revenue?



Certified Fresh vs Rotten

+30%

Revenue premium

Cert. Fresh avg gross

$47M

vs $36M for Rotten films

High-gross films median RT

63%

Fresh Score average

Finding 3 — Critical Reception

How Rotten Tomatoes scores relate to box office revenue

- **Certified Fresh (91–100%)** earns $46.9M avg — approximately +30% more than Rotten films ($36.1M).
- **Not perfectly linear** — the 61–75% Fresh band underperforms due to limited-release prestige films with small theatrical footprints.
- **KDE chart** shows high-grossing films cluster at higher RT scores; low-grossing films concentrate at lower ranges.
- **Low R^2** — blockbuster franchise films earn huge grosses regardless of reviews, but for most films quality matters.
- **Implication:** invest in script development, experienced directors & post-production. Don't cut quality to hit a release date.

Key Insight — Quality Strategy

Strategic implication of the critical reception analysis

Critical reception is a meaningful but not dominant driver of success. Treat quality as a competitive advantage — not a guarantee.

Hire for quality

Directors with strong critical track records for the first 3 films

Invest in scripts

Budget $2M–$5M for script development before greenlighting production

Distribution matters

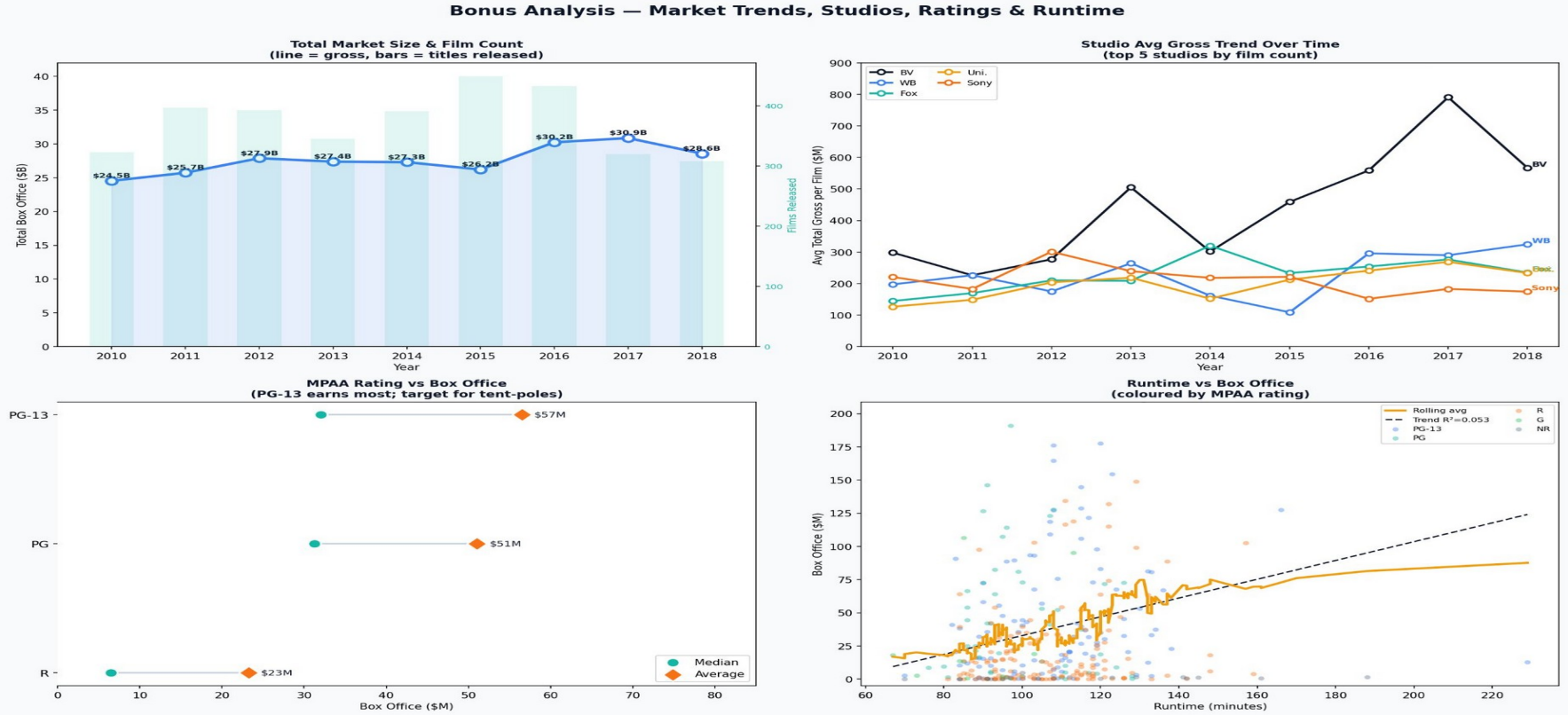
Good scores don't help if the film has a limited release — align scale with quality

Delay over cut

Release date flexibility is a competitive advantage — delay rather than compromise quality

Bonus Analysis — Market Context

Market size trends, studio performance, MPAA ratings, and runtime insights



Our Three Recommendations

Evidence-backed strategies for a successful studio launch

1

Lead with Sci-Fi, Action & Animation

Sci-Fi & Fantasy averages $81M per film — 60% more than Drama. Action & Animation follow at $62M and $56M. These genres dominate worldwide gross and attract the broadest audiences. → Prioritize these three genres in your initial development slate.

2

Budget $100M–$200M per Tent-Pole

Films in the $100–200M budget tier achieve a 93% profit rate — meaning 9 in 10 films recoup their budget — with a median ROI of 2.79×. Larger budgets show higher ROI but require more capital at risk. → Start in this tier and scale as the studio matures.

3

Invest in Quality — Target Certified Fresh

Films scoring above 60% on Rotten Tomatoes earn 30% more revenue than 'Rotten' films. Certified Fresh titles (91–100%) average $47M vs $36M — a clear, measurable premium. → Set a quality threshold: only greenlight projects with a clear path to 60%+ RT.

Detailed Recommendations

Specific actions for each strategic recommendation

1 Lead with Sci-Fi, Action & Animation

Rating target:	PG-13 for max audience breadth ($56.5M avg vs $23.2M for R-rated)
Genres to avoid:	Drama ($28.3M avg), Documentary — too low-margin for a new studio
Secondary play:	Horror micro-budget ($1–5M) for slate diversification & cash flow

2 Budget $100M–$200M per Tent-Pole

Expected returns:	2.79× median ROI = $279M–$558M worldwide on a $100–200M film
Micro-budget option:	$1M–$10M horror films for slate diversification & cash flow
Tier to avoid:	$10M–$50M range: 66% profit rate — worst risk-adjusted tier

3 Invest in Quality — Target Certified Fresh

Hire strategy:	Directors with strong critical track records for first 3 films
Script investment:	Budget $2M–$5M for development before production greenlight
Release policy:	Delay release rather than cut quality — flexibility is a competitive advantage

Final Summary

All objectives, findings and recommendations at a glance

Objective	Key Finding	Statistic	Recommendation
Genre Revenue	Sci-Fi/Fantasy & Action dominate per-film earnings	**$80.9M & $62.3M**	Lead with Sci-Fi, Action, Animation
Budget vs ROI	$100–200M tier has the best risk-adjusted profit rate	**93.1% profit rate**	Target $100M–$200M per tent-pole
Critical Quality	Certified Fresh films earn 30% more than Rotten	**+$10.8M premium**	Invest in quality — Certified Fresh
Market Trend	Market grew 26%; fewer but bigger films each year	**$24.5B → $30.9B**	Release 2–3 large films per year
Studio Benchmark	Disney earns 2× nearest rival per film	**$417M vs $230M**	Model: owned IP, family, quality
MPAA Rating	PG-13 earns the most; R-rated significantly lower	**$56.5M vs $23.2M**	Target PG-13 for all tent-poles

Limitations & Data Notes

Important caveats to consider when interpreting the findings

Nominal figures

All revenue and budget figures are nominal — not adjusted for inflation. Films from 2010 and 2018 are compared in the same dollar terms despite ~18% cumulative inflation over the period.

Small RT sample

The critical reception analysis is based on only 299 matched films — directional and meaningful, but not conclusive at the genre or studio level.

No P&A costs included

Production budgets from The Numbers do not include Prints & Advertising (marketing) spend, which typically equals 50–100% of the production budget. True profitability is lower than the ROI figures suggest.

Genre classification

Genre labels differ between datasets: Rotten Tomatoes uses text tags; TMDB uses numeric IDs. Both are used independently, which may lead to inconsistencies across analyses.

Next Steps

Phase 1

Months 1–3

Development Slate

- Commission 3–5 scripts in Sci-Fi, Action & Animation
- Engage top-tier writers and directors
- Establish quality review with RT score benchmarks

Phase 2

Months 4–9

Budget & Partnerships

- Secure financing for $100–200M tent-pole productions
- Identify studio co-production partners
- Set up production infrastructure and legal entities

Phase 3

Months 10+

Production & Marketing

- Begin production on first approved slate film
- Build marketing strategy targeting PG/PG-13 audience
- Monitor competitive landscape and adjust slate

Q & A

Suggested Discussion Questions

1. Which genre would you prioritize for the studio's first film, and why?
2. How should the studio balance creative risk with the $100M–$200M budget sweet spot?
3. Can a new studio realistically target Certified Fresh status on its first productions?
4. What risks does the analysis not capture that could affect the studio's strategy?

Data: Box Office Mojo · The Numbers · TMDB · Rotten Tomatoes | 2010–2018 Theatrical | Figures nominal